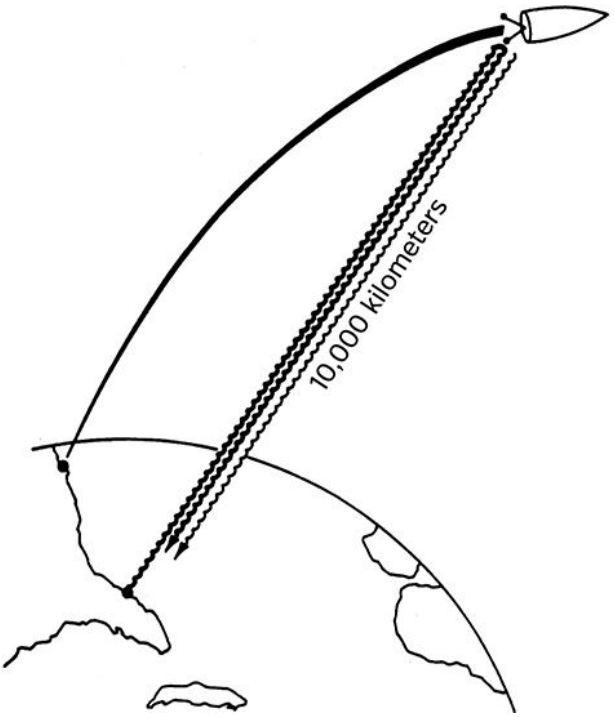


Fig. 4.1. Atomic clocks measure slowing of time on Earth. *[Reproduced from* Was Einstein Right? Putting General Relativity to the Test*, by Clifford M. Will (Basic Books, 1993).]*

The global positioning system (GPS), by which our smart phones can tell us where we are to 10 meters' accuracy, relies on radio signals from a set of 27 satellites at a height of 20,000 kilometers (Figure 4.2). Typically only four to twelve satellites can be seen at once from any location on Earth. Each radio signal from a viewable satellite tells the smart phone where the satellite is located and the time the signal was transmitted. The smart phone measures the signal's arrival time and compares it with its transmission time to learn how far the signal traveled—the distance between satellite and phone. Knowing the locations and distances to several satellites, the smart phone can triangulate to learn its own location.

This scheme would fail if the signal transmission times were the true times measured on the satellite. Time at a 20,000-kilometer height flows more rapidly than on Earth by forty microseconds each day, and the